$H(z)$ **Prijenosna funkcija diskr. sustava**

$u[n] = U z^n \quad , \; z \in \mathbb{C}$

$$y[n] = h[n] * u[n] = \sum_{m=-\infty}^{\infty} h[m] \cdot u[n-m] = \sum_{m=-\infty}^{\infty} h[m] \cdot U \cdot z^{n-m} =$$

$$= U \cdot z^n \underbrace{\sum_{m=-\infty}^{\infty} h[m] \cdot z^{-m}}_{H(z)} \longrightarrow \boxed{H(z) = \sum_{m=-\infty}^{\infty} h[m] \cdot z^{-m}}$$

prijenosna funkcija

$z = e^{j\Omega}$

$$\Longrightarrow H(e^{j\Omega}) = \sum_{m=-\infty}^{\infty} h[m] \cdot e^{-j\Omega m}$$ (opet Fourier)

$\longrightarrow$ Frekv. karakt

$$y[n] = U e^{j\Omega n} \cdot H(e^{j\Omega})$$

$$H(e^{j\omega}) = H(z) \Big|_{z = e^{j\Omega}} \dashrightarrow H(e^{j\Omega}) = |H(e^{j\omega})| \cdot e^{j \measuredangle H(e^{j\Omega})}$$

npr. ako: $u[n] = U \cdot \cos[\Omega n]$

$$y_p[n] = U \cdot |H(e^{j\Omega})| \cdot \cos(\Omega n + \measuredangle H(e^{j\Omega}))$$

Primjer $y[n] - y[n-1] + \frac{1}{2} y[n-2] = u[n]$

1) možemo pretpostaviti da je oblik rješenja pobuda $u[n] = U z^n$

$u[n] = U z^n \quad , \quad y[n] = Y z^n$

$$Y \cancel{z^n} - Y \cancel{z^{n-1}} + \frac{1}{2} Y \cancel{z^{n-2}} = U \cancel{z^n}$$

$$Y - Y z^{-1} + \frac{1}{2} Y z^{-2} = U \longrightarrow Y\left(1 - z^{-1} + \frac{1}{2} z^{-2}\right) = U$$

$$\boxed{Y = U \cdot \underbrace{\frac{1}{1 - z^{-1} + \frac{1}{2} z^{-2}}}_{H(z)}}$$

* Nastavak primjera: $y[n]-y[n-1]+\frac{1}{2}y[n-2]=u[n]$

$$\dots \rightarrow H(z)=\frac{1}{1-z^{-1}+\frac{1}{2}z^{-2}}=\frac{z^2}{z^2-z+\frac{1}{2}}$$

NULE:

$z^2=0 \rightarrow z_{1,2}=0$

POLOVI:

$z^2-z+\frac{1}{2}=0 \longrightarrow z_{p_{1,2}}=\frac{1}{2}\pm\frac{1}{2}j$

! stabilnost: $|z_{p_{1,2}}|=\sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2}=\frac{1}{\sqrt{2}}=\frac{\sqrt{2}}{2}<1$ ✓ sustav stabilan

! DISKR

* kod kont. mora <0

• if stabilan

FREKV. KARAKT: $H(e^{j\Omega})=\dfrac{1}{1-e^{-j\Omega}+\frac{1}{2}e^{-2j\Omega}}$

$$H(e^{j\Omega})=\frac{1}{1-[\cos(-\Omega)-j\sin(-\Omega)]+\frac{1}{2}[\cos(2\Omega)-j\sin(2\Omega)]}$$

$$H(e^{j\Omega})=\frac{1}{1-\cos(-\Omega)+j\sin(-\Omega)+\frac{1}{2}\cos(2\Omega)-\frac{1}{2}j\sin(2\Omega)}$$

$$|H(e^{j\Omega})|=\frac{1}{\sqrt{\left(1-\cos(-\Omega)+\frac{1}{2}\cos(2\Omega)\right)^2+\left(\sin(-\Omega)-\frac{1}{2}\sin(2\Omega)\right)^2}}$$

$$\measuredangle H(e^{j\Omega})=-\operatorname{arctg}\frac{\sin(-\Omega)-\frac{1}{2}\sin(2\Omega)}{1-\cos(-\Omega)+\frac{1}{2}\cos(2\Omega)}$$

u matlabu lako, na papiru teže nacrtati

→ i sad ako kažemo da je pobuda $u[n]=2\cos\left[\frac{\pi}{6}n\right]\mu[n]$

$\Omega=\frac{\pi}{6}$

$$H(e^{j\frac{\pi}{6}})=\frac{1}{1-\cos(\frac{\pi}{6})+j\sin(\frac{\pi}{6})+\frac{1}{2}\cos(\frac{\pi}{3})-\frac{1}{2}j\sin(\frac{\pi}{3})}$$

$$H(e^{j\frac{\pi}{6}})=\frac{1}{0{,}384+j\,0{,}061}$$

$|H(e^{j\pi/6})|=2.56$

$\measuredangle H(e^{j\pi/6})=0.173$

$$\boxed{y_p[n]=2\cdot 2{,}56\cdot\cos\left(\frac{\pi}{6}n-0.173\right)\mu[n]}$$